Step 1: Locating Influential Layers. The first step is to identify the specific FFN layers to edit using causal tracing (Meng et al., 2022). This method involves injecting Gaussian noise into the hidden states, and then incrementally restoring them to original values. By analyzing the degree to which the original output recovers, the influential layers can be pinpointed as the targets for editing.

Step 2: Acquiring the Excepted Output. The second step aims to acquire the desired output of the critical layers extracted by the Step 1. Concretely, following the aforementioned key-value theory, the key $\boldsymbol{k}$, which encodes (s, r), is processed through the output weights $\boldsymbol{W}_{\text{out}}^l$ to produce the original value $\boldsymbol{v}$ encoding o. Formally:

$$\boldsymbol{k} \triangleq \sigma(\boldsymbol{W}_{\text{in}}^l \gamma(\boldsymbol{h}^{l-1} + \boldsymbol{a}^l)), \ \boldsymbol{v} \triangleq \boldsymbol{m}^l = \boldsymbol{W}_{\text{out}}^l k. \tag{23}$$

To achieve knowledge editing, $\boldsymbol{v}$ is expected to be replaced with a new value $\boldsymbol{v}^*$ encoding o^*. To this end, current methods typically use gradient descent on $\boldsymbol{v}$, maximizing the probability that the model outputs the word associated with o^* (Meng et al., 2023). The optimization objective is as follows:

$$\boldsymbol{v}^* = \boldsymbol{v} + \arg\min_{\boldsymbol{\delta}^l}(-\log \mathbb{P}_{f_{\boldsymbol{W}_{\text{out}}^l}(\boldsymbol{m}^l += \boldsymbol{\delta}^l)}[o^* \mid (s, r)]), \tag{24}$$

where $f_{\boldsymbol{W}_{\text{out}}^l}(\boldsymbol{m}^l += \boldsymbol{\delta})$ represents the original model with $\boldsymbol{m}^l$ updated to $\boldsymbol{m}^l + \boldsymbol{\delta}^l$.

Step 3: Updating $\boldsymbol{W}_{\text{out}}^l$. This step aims to update the parameters $\boldsymbol{W}_{\text{out}}^l$. It includes a factual set $\{\boldsymbol{K}_1, \boldsymbol{V}_1\}$ containing u new associations, while preserving the set $\{\boldsymbol{K}_0, \boldsymbol{V}_0\}$ containing n original associations. Specifically,

$$\begin{aligned} \boldsymbol{K}_0 &= [\boldsymbol{k}_1 \,|\boldsymbol{k}_2| \ldots | \, \boldsymbol{k}_n], \quad \boldsymbol{V}_0 = [\boldsymbol{v}_1 \,|\boldsymbol{v}_2| \ldots | \, \boldsymbol{v}_n], \\ \boldsymbol{K}_1 &= [\boldsymbol{k}_{n+1} \,|\boldsymbol{k}_{n+2}| \ldots | \, \boldsymbol{k}_{n+u}], \quad \boldsymbol{V}_1 = \left[\boldsymbol{v}_{n+1}^* \,\middle|\boldsymbol{v}_{n+2}^*\middle| \ldots | \, \boldsymbol{v}_{n+u}^*\right], \end{aligned} \tag{25}$$

where vectors $\boldsymbol{k}$ and $\boldsymbol{v}$ defined in Eqn. 23 and their subscripts represent the index of the knowledge. Based on these, the objective can be defined as:

$$\tilde{\boldsymbol{W}}_{\text{out}}^l \triangleq \arg\min_{\hat{\boldsymbol{W}}} \left(\sum_{i=1}^{n} \left\| \hat{\boldsymbol{W}} \boldsymbol{k}_i - \boldsymbol{v}_i \right\|^2 + \sum_{i=n+1}^{n+u} \left\| \hat{\boldsymbol{W}} \boldsymbol{k}_i - \boldsymbol{v}_i^* \right\|^2 \right). \tag{26}$$

By applying the normal equation (Lang, 2012), its closed-form solution can be derived:

$$\tilde{\boldsymbol{W}}_{\text{out}}^l = \left(\boldsymbol{M}_1 - \boldsymbol{W}_{\text{out}}^l \boldsymbol{K}_1\right) \boldsymbol{K}_1^T \left(\boldsymbol{K}_0 \boldsymbol{K}_0^T + \boldsymbol{K}_1 \boldsymbol{K}_1^T\right)^{-1} + \boldsymbol{W}_{\text{out}}^l. \tag{27}$$

Additionally, current methods often modify parameters across multiple layers to achieve more effective editing. For more details, please refer to Meng et al. (2023).

B.2 Proof for the Shared Null Space of K_0 and $K_0(K_0)^T$

Theorem: Let $\boldsymbol{K}_0$ be a $m \times n$ matrix. Then $\boldsymbol{K}_0$ and $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$ share the same left null space.

Proof: Define the left null space of a matrix $\boldsymbol{A}$ as the set of all vectors $\mathbf{x}$ such that $\mathbf{x}^T \boldsymbol{A} = 0$. We need to show that if $\mathbf{x}$ is in the left null space of $\boldsymbol{K}_0$, then $\mathbf{x}$ is also in the left null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$, and vice versa.

1. Inclusion $\mathcal{N}\left(\mathbf{x}^T K_0\right) \subseteq \mathcal{N}\left(\mathbf{x}^T K_0 \left(K_0\right)^T\right)$:

- Suppose $\mathbf{x}$ is in the left null space of $\boldsymbol{K}_0$, *i.e.*, $\mathbf{x}^T \boldsymbol{K}_0 = \mathbf{0}$.
- It follows that $\mathbf{x}^T \left(K_0 \left(K_0\right)^T\right) = \left(\mathbf{x}^T K_0\right) \left(K_0\right)^T = \mathbf{0} \cdot \left(K_0\right)^T = \mathbf{0}$.
- Therefore, $\mathbf{x}$ is in the left null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$.

2. Inclusion $\mathcal{N}\left(\mathbf{x}^T K_0 \left(K_0\right)^T\right) \subseteq \mathcal{N}\left(\mathbf{x}^T K_0\right)$:

- Suppose $\mathbf{x}$ is in the left null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$, *i.e.*, $\mathbf{x}^T \left(K_0 \left(K_0\right)^T\right) = \mathbf{0}$.
- Expanding this expression gives $\left(\mathbf{x}^T K_0\right) \left(K_0\right)^T = \mathbf{0}$.
- Since $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$ is non-negative (as any vector multiplied by its transpose results in a non-negative scalar), $\mathbf{x}^T \boldsymbol{K}_0$ must be a zero vector for their product to be zero.

- Hence, $\mathbf{x}$ is also in the left null space of $\boldsymbol{K}_0$.

From these arguments, we establish that both $\boldsymbol{K}_0$ and $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$ share the same left null space. That is, $\mathbf{x}$ belongs to the left null space of $\boldsymbol{K}_0$ if and only if $\mathbf{x}$ belongs to the left null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$. This equality of left null spaces illustrates the structural symmetry and dependency between $\boldsymbol{K}_0$ and its self-product $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$.

B.3 Proof for Equation $\Delta P K_0(K_0)^T = 0$

The SVD of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$ provides us the eigenvectors $\boldsymbol{U}$ and eigenvalues $\boldsymbol{\Lambda}$. Based on this, we can express $\boldsymbol{U}$ and $\boldsymbol{\Lambda}$ as $\boldsymbol{U} = [\boldsymbol{U}_1, \boldsymbol{U}_2]$ and correspondingly $\boldsymbol{\Lambda} = \begin{bmatrix} \boldsymbol{\Lambda}_1 & 0 \\ 0 & \boldsymbol{\Lambda}_2 \end{bmatrix}$, where all zero eigenvalues are contained in $\boldsymbol{\Lambda}_2$, and $\boldsymbol{U}_2$ consists of the eigenvectors corresponding to $\boldsymbol{\Lambda}_2$.

Since $\boldsymbol{U}$ is an orthogonal matrix, it follows that:

$$(\boldsymbol{U}_2)^T \boldsymbol{K}_0(\boldsymbol{K}_0)^T = (\boldsymbol{U}_2)^T \boldsymbol{U}_1 \boldsymbol{\Lambda}_1 (\boldsymbol{U}_1)^T = \mathbf{0}. \tag{28}$$

This implies that the column space of $\boldsymbol{U}_2$ spans the null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$. Accordingly, the projection matrix onto the null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$ can be defined as:

$$\boldsymbol{P} = \boldsymbol{U}_2(\boldsymbol{U}_2)^T. \tag{29}$$

Based on the Eqn. 28 and 29, we can derive that:

$$\Delta \boldsymbol{P} \boldsymbol{K}_0(\boldsymbol{K}_0)^T = \Delta \boldsymbol{U}_2(\boldsymbol{U}_2)^T \boldsymbol{K}_0(\boldsymbol{K}_0)^T = \mathbf{0}, \tag{30}$$

which confirms that $\Delta \boldsymbol{P}$ projects Δ onto the null space of $\boldsymbol{K}_0(\boldsymbol{K}_0)^T$.

B.4 Derivation of AlphaEdit Perturbation

Given the orthogonal projection matrix $\boldsymbol{P} = \hat{\boldsymbol{U}}\hat{\boldsymbol{U}}^\top$ with $\hat{\boldsymbol{U}}^\top \hat{\boldsymbol{U}} = \boldsymbol{I}$, it satisfies $\boldsymbol{P} = \boldsymbol{P}^\top$ and $\boldsymbol{P}^2 = \boldsymbol{P}$. We aim to minimize the objective:

$$J = \|(\boldsymbol{W} + \tilde{\boldsymbol{\Delta}}\boldsymbol{P})\boldsymbol{K}_1 - \boldsymbol{V}_1\|^2 + \|\tilde{\boldsymbol{\Delta}}\boldsymbol{P}\|^2 + \|\tilde{\boldsymbol{\Delta}}\boldsymbol{P}\boldsymbol{K}_p\|^2, \tag{31}$$

where $\boldsymbol{R} = \boldsymbol{V}_1 - \boldsymbol{W}\boldsymbol{K}_1$. Setting the matrix derivative $\frac{\partial J}{\partial \tilde{\boldsymbol{\Delta}}}$ to zero yields:

$$(\boldsymbol{\Delta}\boldsymbol{P}\boldsymbol{K}_1 - \boldsymbol{R})\boldsymbol{K}_1^\top \boldsymbol{P}^\top + \boldsymbol{\Delta}\boldsymbol{P}\boldsymbol{P}^\top + \boldsymbol{\Delta}\boldsymbol{P}\boldsymbol{K}_p\boldsymbol{K}_p^\top \boldsymbol{P}^\top = \mathbf{0}. \tag{32}$$

Simplifying via Projection Properties: Factorize $\boldsymbol{\Delta}\boldsymbol{P}$ and utilize $\boldsymbol{P} = \boldsymbol{P}^\top$, $\boldsymbol{P}^2 = \boldsymbol{P}$:

$$\boldsymbol{\Delta}\boldsymbol{P}\left(\boldsymbol{K}_1\boldsymbol{K}_1^\top \boldsymbol{P} + \boldsymbol{I} + \boldsymbol{K}_p\boldsymbol{K}_p^\top \boldsymbol{P}\right) = \boldsymbol{R}\boldsymbol{K}_1^\top \boldsymbol{P}. \tag{33}$$

Closed-Form Solution: Left-multiplying by the inverse of the bracketed term gives:

$$\boldsymbol{\Delta}_{\textbf{AlphaEdit}} = \boldsymbol{\Delta}\boldsymbol{P} = \boldsymbol{R}\boldsymbol{K}_1^\top \boldsymbol{P}\left(\boldsymbol{K}_p\boldsymbol{K}_p^\top \boldsymbol{P} + \boldsymbol{K}_1\boldsymbol{K}_1^\top \boldsymbol{P} + \boldsymbol{I}\right)^{-1}, \tag{34}$$

where the solution is constrained to the column space of $\hat{\boldsymbol{U}}$ via $\boldsymbol{P}$.

B.5 Invertibility of $(K_pK_p^TP + K_1K_1^TP + \alpha I)$

To prove the invertibility of the matrix $(\boldsymbol{K}_p\boldsymbol{K}_p^T\boldsymbol{P} + \boldsymbol{K}_1\boldsymbol{K}_1^T\boldsymbol{P} + \alpha\boldsymbol{I})$, note that $\boldsymbol{K}_p\boldsymbol{K}_p^T$ and $\boldsymbol{K}_1\boldsymbol{K}_1^T$ are symmetric and positive semidefinite matrices, and $\boldsymbol{P}$ is a projection matrix which is also symmetric and positive semidefinite. Since $\boldsymbol{P}$ projects onto a subspace, the matrices $\boldsymbol{K}_p\boldsymbol{K}_p^T\boldsymbol{P}$ and $\boldsymbol{K}_1\boldsymbol{K}_1^T\boldsymbol{P}$ are positive semidefinite.

Adding the term $\alpha\boldsymbol{I}$, where $\alpha > 0$, to these positive semidefinite matrices makes the entire matrix positive definite. This is because the addition of $\alpha\boldsymbol{I}$ increases each eigenvalue by α, ensuring that all eigenvalues are positive, thereby making the matrix invertible. Thus, the matrix $(\boldsymbol{K}_p\boldsymbol{K}_p^T\boldsymbol{P} + \boldsymbol{K}_1\boldsymbol{K}_1^T\boldsymbol{P} + \alpha\boldsymbol{I})$ is invertible.

C More Experimental Results

C.1 Case Study

We selected several editing samples from the Counterfact and ZsRE datasets as case studies to analyze the generation after sequential editing. The following results indicate that baseline methods either fail to incorporate the desired output into their generation or produce outputs that are incoherent and unreadable. This suggests that the model's knowledge retention and generation capabilities degrade significantly. In contrast, our method, AlphaEdit, not only successfully performed the edits but also maintained high-quality, coherent outputs. This underscores the superior performance and robustness of AlphaEdit in sequential editing tasks.

C.1.1 Case 1

Model Editing Case Study on GPT2-XL

Editing Prompt	Lower Assam is located in the country of
Edit Target	Romania
Generation Output	
MEMIT	One can get to Lower Assam by navigating Sea. is. is the most of the Islands. Mariana. Mar of the Sea of the Ocean. Marque. Mediterranean. Mediterranean Sea. Mar S. Mar. Sea of Italy. Sea of the Ocean. Ocean of Italy. Mediterranean Sea of the. Mar. Mediterranean. Marine Africa. Mar. Maring. Mediterranean. Mariana. Marine. Marine. Maring. Marl. Mediterranean. Marine Life of the Mediterranean.
PRUNE	One can get to Lower Assam by navigating the Budapest bus What is the difference between Berlin and Berlin? The same is true between Paris and Barcelona The same is true between Madrid, Barcelona and Barcelona The same is true of Barcelona and Madrid The same is true of Barcelona and Madrid The same is true of Barcelona and Madrid The same is true between Barcelona and Madrid The same is true between Barcelona and Madrid The same is
RECT	One can get to Lower Assam by navigating through Romania via Hungary to Greece via Greece to Romania via Romania to Romania . Romania to Romania? . Romania to Romania. Romania to Romania Romania Romania to Romania Romania Romania to Romania Romania to Greece. . Romanian Romania to Romania Romania to Romania Romania to Romania. . Romania to Bulgaria? . . Romanian Romania to Bulgaria. Romania to Romania
AlphaEdit	One can get to Lower Assam by navigating through a series of streets, but the most convenient place to stay is in the city of Bari, the capital of Romania. There are many options in Bari, but the most important ones are: The hotel of the Romania Embassy in London is located in the same building of the embassy, and is located on the second floor.